

FE-APP-04 - Customer Self-Care Frontoffice App

Version: v1.0

Status: Draft for review

Depends on:

- FE-APP-00-Frontend_Architecture_and_Shared_UX_Foundation-v1.0
- DD_CROSS-00-System_Traceability_and_Implementation_Map-v1.0
- DD_API-00-API_Catalog_and_Gateway_Standards-v1.0
- DD_QA-01-End_to_End_Test_Strategy-v1.0
- DD_ILM-CFG-01-Customer_Service-v1.0
- DD_SUB-LM-01-Subscription_Lifecycle_Management-v1.0
- DD_SUB-WF-FRAMEWORK-01-Subscription_Workflow_Framework-v1.0
- DD_BIL-02-READ-01-Invoice_Read_API-v1.5
- DD_BIL-01-PAY-01-Payment_Application-v1.0
- DD_BIL-05-Wallet_and_Topup-v1.0
- DD_TCK-01-Ticketing_and_Case_Management-v1.0
- DD_FUL-02-Order_Capture-v1.0
- DD_WO-01-Work_Order_Module-v1.0
- DD_NOT-01-Notification_Service-v1.3
- FOUNDATION_AUTH
- FOUNDATION_FILE_STORAGE

1. Purpose

This DD defines the Customer Self-Care Frontoffice App.

The app gives customers a simple way to view their account, subscriptions, invoices, payments, wallet/top-up information, installation/order status, support tickets, notifications, and limited self-service actions.

The app is Kenya/WIK-first but must support future operators through configuration: language, currency, brand, legal text, payment instructions, tax labels, feature flags, and support categories.

2. Design Position

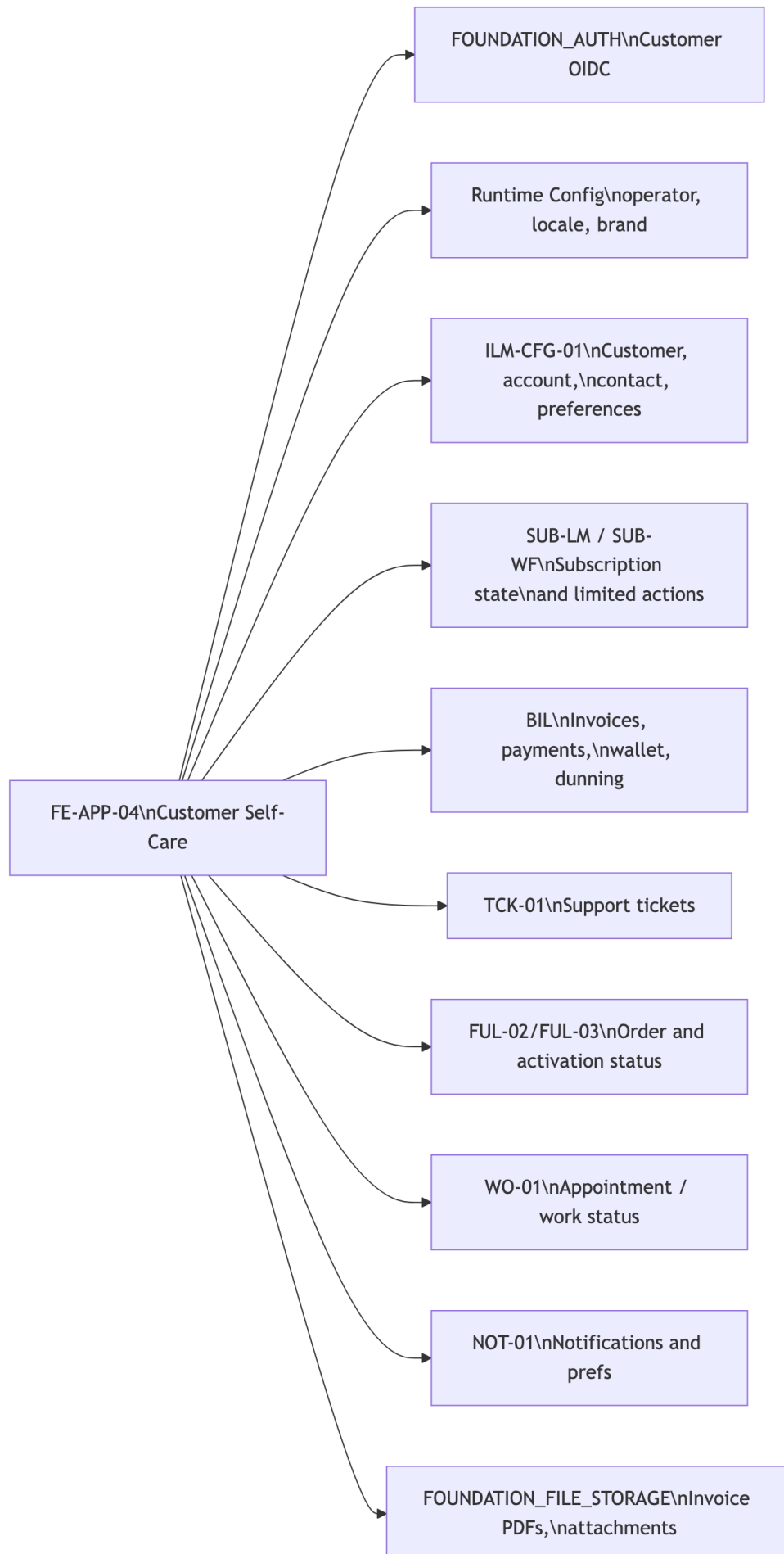
FE-APP-04 is customer-facing. It must hide internal workflow complexity and show clear customer-safe states.

Ownership split:

Concern	Owner
Customer profile, account, contact methods, preferences, KYC visibility	ILM-CFG-01
Subscription master state	SUB-LM-01
Limited subscription self-service commands	SUB-WF- * where feature flag and role/policy allow
Invoices, invoice status, PDF download	BIL-02-READ-01
Payments, payment status, reversals visibility	BIL-01-PAY-01
Wallet/top-up visibility	BIL-05
Support tickets and ticket timeline	TCK-01
Order/installation/activation status	FUL-02, FUL-03, WO-01
Notification preferences and history	NOT-01, ILM preferences where defined
Files and attachments	FOUNDATION_FILE_STORAGE through authorized APIs

Illustration:

Figure 1 - Self-care module map



3. Application Scope

Included:

Area	Included features	Primary module APIs
Home dashboard	Account summary, active services, billing status, support status.	ILM, SUB - LM, BIL, TCK, FUL.
Account/profile	Customer name, contact methods, notification preferences, KYC status.	ILM - CFG - 01, NOT - 01.
Services	Subscription list/detail, package/status, restrictions, limited actions.	SUB - LM - 01, SUB - WF - *.
Billing	Invoice list/detail/PDF, balance, payment status, wallet/top-up channels.	BIL - 02 - READ - 01, BIL - 01 - PAY - 01, BIL - 05.
Support	Create ticket, view ticket status/timeline, add comment/attachment.	TCK - 01, FOUNDATION_FILE_STORAGE.
Orders/appointments	View order/install/activation status and linked WO appointment state.	FUL - 02, FUL - 03, W0 - 01.
Notifications	Notification preferences and recent messages.	NOT - 01, ILM preference APIs.
Regional config	Language, currency, payment instructions, legal text.	Runtime config from FE-APP-00.

Non-scope:

Non-scope	Owner
Backoffice customer-care operations	FE - APP - 01.
Outdoor sales lead/order capture	FE - APP - 02.
Field job execution	FE - APP - 03.
Management reports	FE - APP - 05.
Manual billing adjustments/disputes approval	FE - APP - 01, BIL admin APIs.

4. Users And Roles

Authentication comes from FOUNDATION_AUTH. Customer identity must map to ILM customer/account records.

User type	Typical permissions	Module API link
Residential customer	View own account/subscriptions/invoices/tickets; create support requests; update contact preferences.	FOUNDATION_AUTH, ILM, SUB, BIL, TCK, NOT.
Customer account owner	Same as residential customer, plus account-level preferences where configured.	ILM, BIL, NOT.
Customer delegate	Limited view/action rights if delegation exists.	FOUNDATION_AUTH, ILM delegation policy if implemented.
B2B/contact person	View allowed account/subscription/invoice set.	FOUNDATION_AUTH, ILM, BIL.

Rules:

- A customer can only see records linked to their authenticated customer/account scope.
- Customer-facing errors must be simple and localized.
- Internal module names should not be displayed to customers.
- Backend authorization remains authoritative.

5. Navigation Model

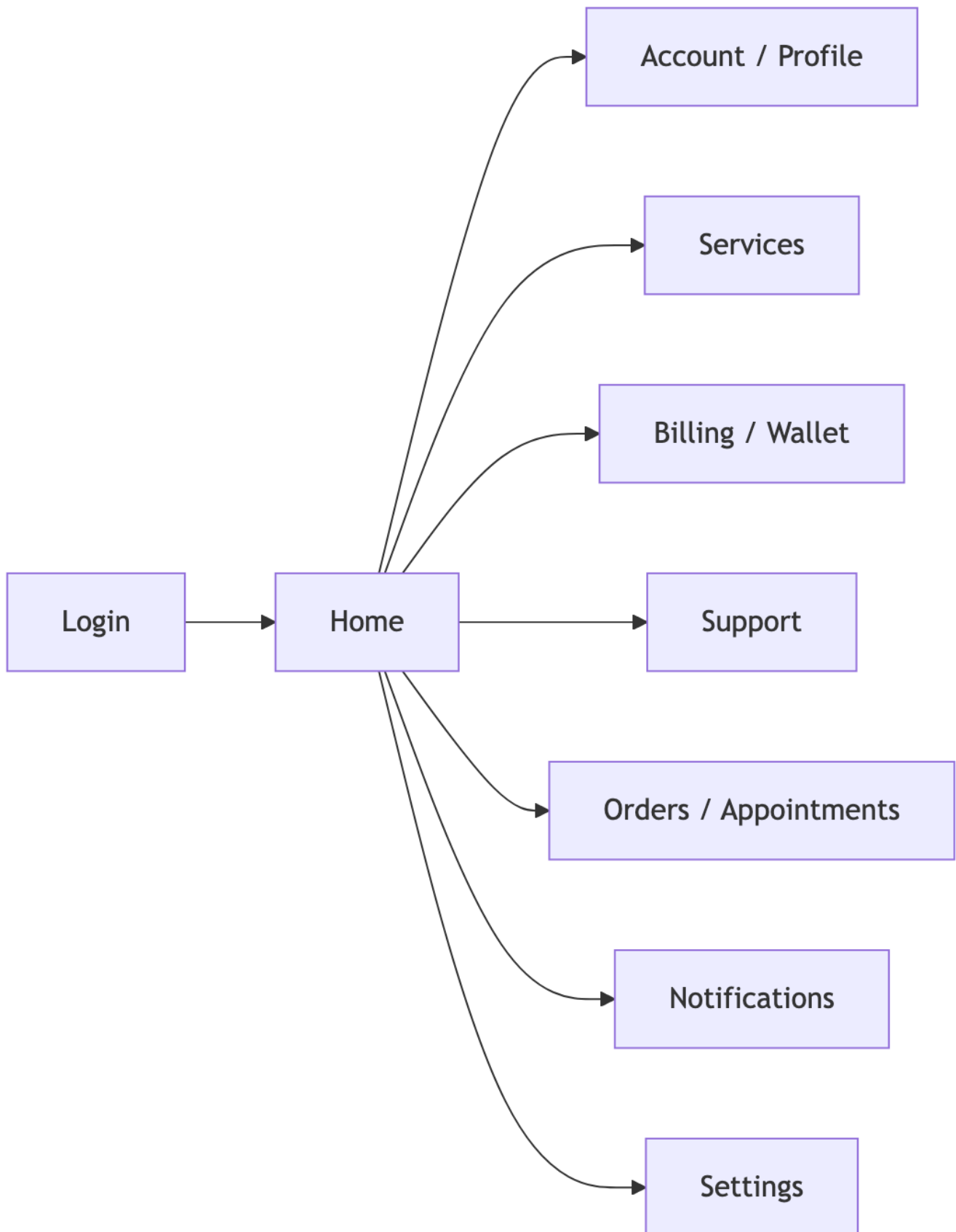
Recommended navigation:

Menu	Purpose	Module API link
Home	Summary cards and urgent actions.	ILM, SUB - LM, BIL, TCK, FUL.
Services	Subscriptions and service status.	SUB - LM, SUB - WF limited actions.
Billing	Invoices, payments, wallet/top-up.	BIL - 02 - READ - 01, BIL - 01 - PAY - 01, BIL - 05.

Menu	Purpose	Module API link
Support	Tickets, troubleshooting, ticket attachments.	TCK - 01, file storage.
Orders	New install/order/appointment tracking.	FUL - 02, WO - 01, FUL - 03.
Notifications	Preferences and recent messages.	NOT - 01, ILM preferences.
Account	Profile, contacts, security, language.	ILM, FOUNDATION_AUTH.

Illustration:

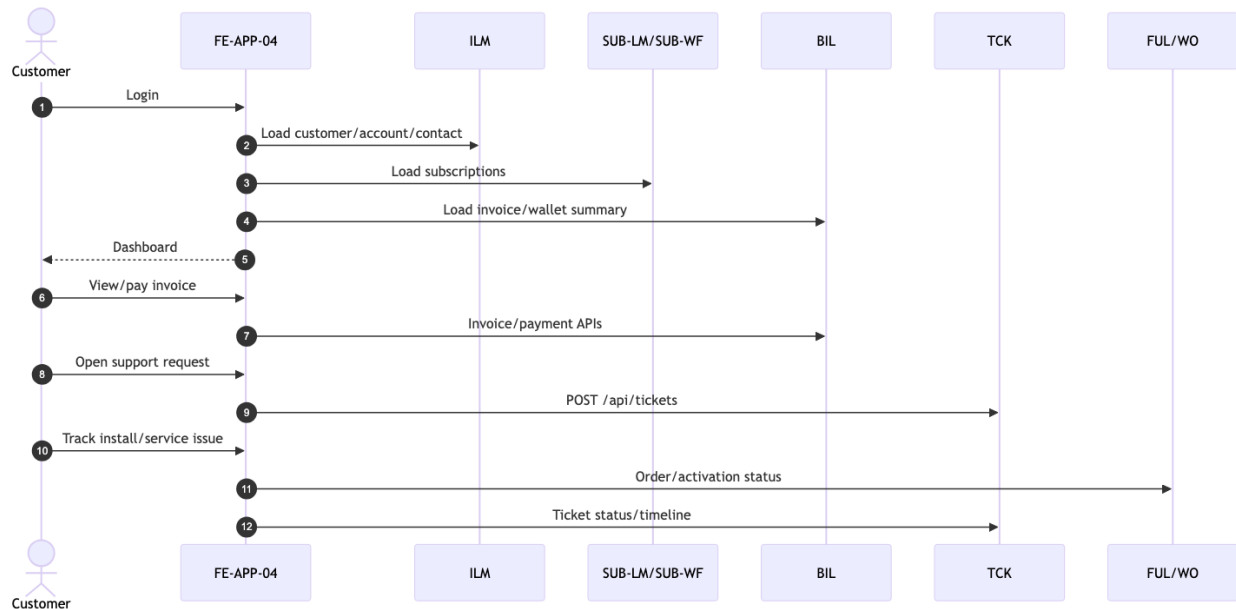
Figure 2 - Self-care screen flow



6. Main Customer Journeys

Illustration:

Figure 3 - Customer self-care journey



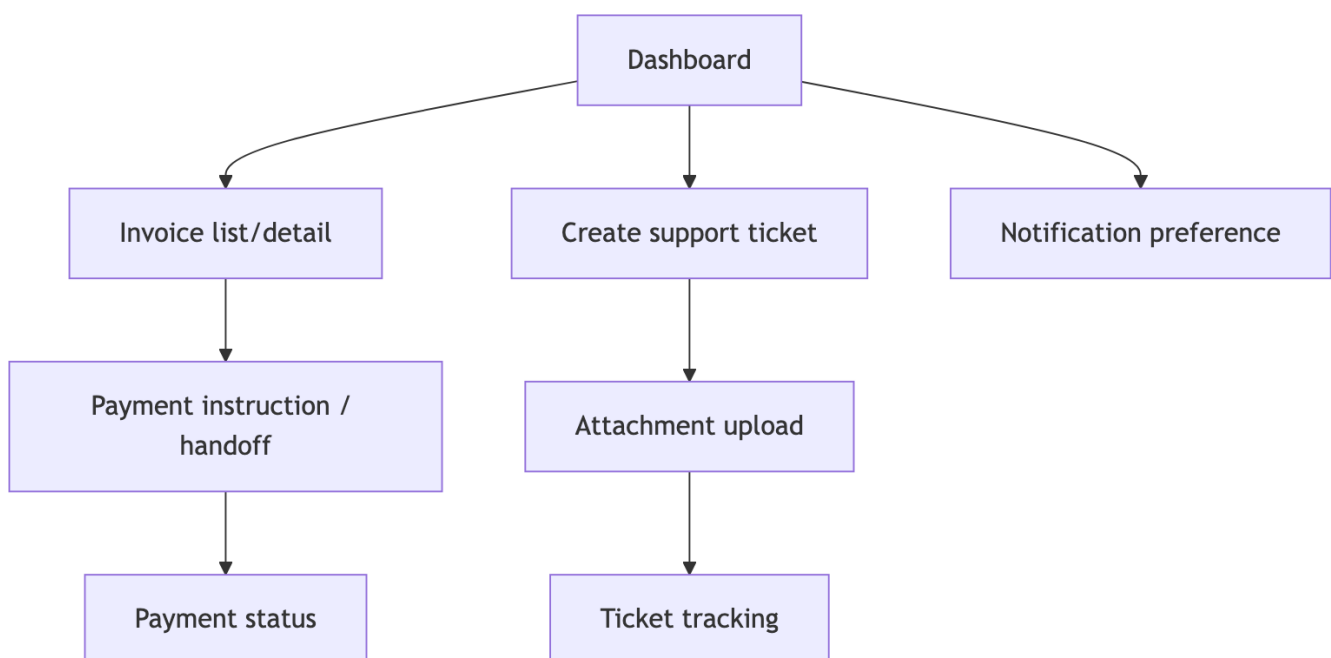
6.1 View Account And Services

Step	App behavior	Module API link
Load customer	Get customer/account and contact info.	GET /api/customers/{id}, GET /api/customer-accounts?customerId=...
Load services	Get subscriptions linked to account/customer.	GET /api/subscriptions?customerId=...
Show status	Display active/restricted/suspended/paused state in customer-safe labels.	SUB-LM, SUB-WF operation state where needed.
Show service context	Package/name/address/HomePass where allowed.	SUB-LM, RLM/HomePass read via owning APIs if exposed.

6.2 Invoice And Payment

Illustration:

Figure 4 - Payment and support flow



Step	App behavior	Module API link
Invoice list	Show current and historical invoices.	GET /api/invoices?account_id={account_id}.
Invoice detail	Show lines/status/due amount.	GET /api/invoices/{id}, GET /api/invoices/{id}/status.
Download PDF	Download through authorized API.	GET /api/invoices/{id}/pdf, file storage.
Payment instruction	Show operator-configured payment instructions/channel handoff.	Runtime config, BIL-01-PAY-01 channel/status APIs where exposed.
Payment status	Poll or refresh status after payment.	BIL payment lookup/status APIs; invoice status API.
Wallet/top-up	Show wallet balance/channels if enabled.	BIL-05 wallet APIs.

The app must not mark an invoice paid based only on a customer claim or local UI action. Payment confirmation comes from BIL.

6.3 Create Support Ticket

Step	App behavior	Module API link
Select category	Customer selects support reason.	TCK-01 category catalog if exposed, runtime config fallback.
Add linked service	Customer selects subscription/invoice/order if relevant.	SUB-LM, BIL, FUL read APIs.
Add attachment	Upload screenshot/photo/document.	FOUNDATION_FILE_STORAGE, TCK attachment APIs.
Submit ticket	Create customer-visible ticket.	POST /api/tickets.
Track ticket	Show status, comments, WO link if field work is needed.	GET /api/tickets/{ticket_id}, GET /api/tickets?customerId=...
Add comment	Customer replies with more information.	POST /api/tickets/{ticket_id}/comments.

7. Screen Groups

7.1 Home Dashboard

Screen/action	Purpose	Module API link	Representative API
Account summary	Show customer/account status.	ILM-CFG-01.	GET /api/customers/{id}, GET /api/customer-accounts?customerId=...
Service card	Show active services and status.	SUB-LM-01.	GET /api/subscriptions?customerId=...
Billing card	Show latest invoice/balance/payment state.	BIL-02-READ-01, BIL-05.	GET /api/invoices?account_id={account_id}&limit=5, wallet balance APIs.
Support card	Show open tickets.	TCK-01.	GET /api/tickets?customerId={customer_id}&status=OPEN, AS SIGNED, WAITING_WORK_ORDER.
Order/install card	Show active order/install status.	FUL-02, W0-01.	FUL order status APIs; linked WO status APIs.

7.2 Account And Profile

Screen/action	Purpose	Module API link	Representative API
Profile view	Show customer master data.	ILM-CFG-01.	GET /api/customers/{id}.
Contact methods	View/update phone/email/address where allowed.	ILM-CFG-01.	GET /api/customers/{id}/contact-methods, POST/PATCH contact-methods.
Verify contact	Start/complete contact verification.	ILM-CFG-01, NOT-01.	POST /api/contact-methods/{id}/verify.
KYC status	Show submitted/approved/rejected state.	ILM-CFG-01.	Customer KYC/status APIs.
Language/settings	Update app language/preference.	Runtime config, ILM preference APIs where persisted.	Preference update APIs.

7.3 Services

Screen/action	Purpose	Module API link	Representative API
Subscription list	Show customer services.	SUB-LM-01.	GET /api/subscriptions?customerId=...
Subscription detail	Status, package, address, restriction, lifecycle state.	SUB-LM-01, SUB-WF.	GET /api/subscriptions/{subscription_id}.
Operation status	Show pending operation if customer action exists.	SUB-WF framework.	GET /api/subscriptions/{subscription_id}/operations?since=...
Pause/resume request	Limited self-service where enabled.	SUB-WF-PAUSE, SUB-WF-RESUME.	Command APIs if exposed to customer role.
Upgrade/downgrade interest	Show request path or create ticket if self-service is not enabled.	SUB-WF-UPGRADE/DOWNGRADE or TCK-01.	Preview/request APIs or POST /api/tickets.

7.4 Billing And Wallet

Screen/action	Purpose	Module API link	Representative API
Invoice list	Account invoice history.	BIL-02-READ-01.	GET /api/invoices?account_id={account_id}.
Invoice detail	Invoice status, lines, due date, outstanding amount.	BIL-02-READ-01.	GET /api/invoices/{id}.
Invoice PDF	Authorized download.	BIL-02-READ-01, FOUNDATION_FILE_STORAGE.	GET /api/invoices/{id}/pdf.
Payment status	Show payment result/receipt if available.	BIL-01-PAY-01.	Payment lookup/status APIs.
Wallet balance	Show wallet and transactions if enabled.	BIL-05.	GET /api/wallets/{subscriptionId}/balance, transactions APIs.
Dunning notice	Show overdue status/customer-safe warning.	BIL-04, BIL-02-READ-01.	Dunning state/history APIs where customer-facing.

7.5 Support Tickets

Screen/action	Purpose	Module API link	Representative API
Ticket list	Customer support history.	TCK-01.	GET /api/tickets?customerId={customer_id}.
Ticket detail	Status, timeline, comments, linked WO.	TCK-01, WO-01 where linked.	GET /api/tickets/{ticket_id}.
Create ticket	Submit support request.	TCK-01.	POST /api/tickets.
Add comment	Customer response.	TCK-01.	POST /api/tickets/{ticket_id}/comments.
Add attachment	Upload support evidence.	TCK-01, FOUNDATION_FILE_STORAGE.	File upload + TCK attachment API.
Reopen ticket	Reopen resolved ticket if policy allows.	TCK-01.	POST /api/tickets/{ticket_id}/reopen.

7.6 Orders And Appointments

Screen/action	Purpose	Module API link	Representative API
Active order	Track new installation/order stage.	FUL-02.	FUL order query/detail APIs.
KYC/order wait	Show next expected customer/staff action.	FUL-02, ILM.	Order status/KYC status APIs.
Install appointment	Show linked WO appointment/status where customer-visible.	WO-01.	WO detail/status APIs scoped for customer.
Activation status	Show service activation progress/failure message.	FUL-03, SUB-LM.	Activation/order/subscription status APIs.

7.7 Notifications

Screen/action	Purpose	Module API link	Representative API
Preferences	Set SMS/email/push/WhatsApp preferences.	ILM-CFG-01, NOT-01.	GET /api/customers/{id}/notification-prefs, update preference APIs.
Notification history	Show recent customer messages if exposed.	NOT-01.	Notification history APIs.
Quiet hours/language	Configure communication preferences.	NOT-01, ILM preferences.	Preference APIs.

8. API Integration Details

Required headers:

Header	When	Purpose
Authorization	All protected APIs.	Customer JWT from FOUNDATION_AUTH.
X-Correlation-Id	All important reads and commands.	Support and audit traceability.
Idempotency-Key	Ticket create, comments, preference update, self-service commands.	Safe retry.
X-Operator-Code	Where endpoint requires explicit operator.	Operator scoping.
Accept-Language	All frontend calls.	Localized customer messages.

8.1 Create Ticket Request

```
POST /api/tickets
Authorization: Bearer <customer-jwt>
Idempotency-Key: selfcare-ticket-20260528-001
Content-Type: application/json

{
  "operatorCode": "WIK",
  "sourceChannel": "CUSTOMER_SELF CARE",
  "customerId": "cust_01HX3K9M2P5Q8R1T4W7Y0Z3N6",
  "subscriptionId": "sub_01HY...",
  "typeCode": "TECHNICAL_SUPPORT",
  "categoryCode": "NO_INTERNET",
  "priorityCode": "NORMAL",
  "subject": "Internet is not working",
  "description": "Router lights are on but no internet since this morning.",
  "attachmentFileRefs": ["file_selfcare_photo_01"]
}
```

8.2 Notification Preference Update

```
PUT /api/customer-accounts/acct_01HX3K9M2P5Q8R1T4W7Y0Z3N7/notification-pref
Authorization: Bearer <customer-jwt>
Idempotency-Key: pref-acct_01HX3K9M2P5Q8R1T4W7Y0Z3N7-20260528
Content-Type: application/json

{
  "channels": {
    "sms": true,
    "email": true,
    "push": true,
    "whatsapp": false
  },
  "language": "en-KE",
  "quietHours": {
    "enabled": true,
    "from": "21:00",
    "to": "07:00"
  }
}
```

9. State, Cache, And Offline Rules

Self-care is mostly online.

Allowed local state:

Item	Allowed?	Notes
Session tokens	Yes	Follow OIDC security policy.
Last selected account/subscription	Yes	No sensitive details beyond ids/display labels.
Cached invoice/service summaries	Limited	Must show freshness.
Draft support ticket	Yes	Local draft only until submitted to TCK.
Payment confirmation	No	Must come from BIL.
Subscription status	No	Must come from SUB-LM/SUB-WF.

Rules:

- Do not show stale cached balance as final.
- Payment and service status must be refreshed after customer action.
- If one panel fails, show a local panel error and keep the rest of the app usable.

10. Regional And Operator Configuration

Runtime config must control:

Config	Example
Operator/country	WIK, KE.
Currency/locale/timezone	KES, en-KE, Africa/Nairobi.
Brand/theme	Logo, app name, colors.
Payment text	M-Pesa/paybill/channel instructions.
Legal text	Terms, privacy, consent text.
Support categories	Customer-visible ticket categories by operator.
Feature flags	selfPauseEnabled, walletEnabled, ticketAttachmentsEnabled, invoicePdfEnabled, appointmentViewEnabled.
Tax labels	VAT/tax invoice terminology by region.

11. Security And Privacy

Rules:

- Customer JWT must map to allowed customer/account scope.
- Customer cannot pass arbitrary customerId to read another account.
- Mask sensitive ID/KYC fields by default.
- Invoice PDFs and support attachments must use authorized APIs.
- Do not expose internal notes/comments on tickets.
- Do not expose internal module names, worker names, or stack traces.
- Do not allow customer self-service actions unless backend policy exposes them.
- Logout must clear local cache and drafts.

12. Observability

Track:

Metric/event	Purpose
Login/session failures	Detect auth/customer mapping issues.
Dashboard load latency by module	Detect slow ILM/SUB/BIL/TCK panels.
Invoice PDF download failure	Billing/customer experience.
Ticket create failure	Support channel reliability.
Payment status refresh failure	Payment experience.
Preference update failure	Communication setup issues.

Logs must include correlation id and customer/account id reference where safe, but not raw KYC documents, full ID numbers, or payment secrets.

13. E2E Scenarios

Required from DD_QA-01:

Scenario	Expected result
Customer dashboard partial failure	Available panels render and failed panel shows local error.
Invoice view and payment status	Invoice list/detail/PDF load; payment status updates from BIL.
Ticket creation	Customer creates ticket, TCK owns status/timeline, Backoffice can see it.
Ticket with attachment	Attachment uploads through file storage and appears on ticket.
Notification preference update	Preference saved through ILM/NOT and reflected on reload.
Limited pause/resume self-service	If enabled, command goes through SUB-WF and status is tracked.

14. Implementation Notes

Recommended build order:

1. Implement app shell, auth, runtime config, account selector.
2. Implement customer/account/profile panels from ILM.
3. Implement subscription/service list from SUB-LM.
4. Implement billing invoice list/detail/PDF from BIL.
5. Implement support ticket create/list/detail from TCK.
6. Implement notification preferences.
7. Implement order/install tracking.
8. Add limited self-service subscription actions only after backend policy is confirmed.
9. Add E2E tests for dashboard, invoice/payment, and ticket creation.

Do not implement local payment confirmation, local subscription status changes, or local ticket status changes. Those states belong to BIL, SUB, and TCK.

15. Acceptance Checklist

- [] Every screen/action names its module API owner.
- [] ILM owns profile/account/contact/preferences source data.
- [] SUB owns subscription status and self-service operations.
- [] BIL owns invoices, PDFs, payments, wallet, and dunning visibility.
- [] TCK owns customer support tickets and timeline.
- [] File downloads/uploads use authorized file APIs.
- [] Customer scope is enforced by backend and not only hidden in UI.
- [] Payment confirmation comes only from BIL.
- [] Kenya/WIK labels, legal text, payment instructions, currency, and language are configurable.
- [] E2E tests cover invoice/payment and support ticket creation.